

eCH-1234 Template Quarto Document

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Abstract Brief summary of the purpose of the document.

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Note

This document uses a gender-neutral formulation when referring to persons. This is based on the guidelines (German) of the Federal Chancellery. Depending on the situation, paired forms

(citizens), gender-abstract forms (insured person), gender-neutral forms (insured person) or paraphrases without personal reference are used. The generic masculine (citizen) is not permitted. Full forms are used in continuous texts, i.e. in texts consisting of formulated sentences. Short forms can be used in abbreviated text passages, namely in tables. The short form is used with a slash but without an ellipsis (referent). Gender asterisks and similar spellings are not used.

1 Introduction

1.1 Status

Approved: This document was approved by the Experts' Committee. It has normative power for the defined field of application in the determined scope of application.

1.2 Scope of application

The information in this chapter should provide the reader with a brief overview of what this standard is intended for. Information about the following matters may be helpful here.

1.3 We can have sub-headings

And write some text.

1.3.a Also sub-sub-headings

And write some more text. Maybe even with a pretty image.



Figure 1: Always add some text to describe what the image shows.

When displaying diagrams, try to write them in Mermaid JS straight away; this makes changes in the future or translations straightforward.

2 Technical notes

We use the ROBOT CLI in our project [1], especially for it's ability to run the HermiT reasoner [2].

3 Data Model

3.1 Genre

A musical category in the Chinook dataset.

Target Class: `:Genre`

Table 1: Genre properties

Description	Path	Type	Cardinality
Name	<code>schema:name</code>		1..1
Part of	<code>schema:partOf</code>	<code>:Genre</code> or <code>sh:IRI</code>	0..*

3.2 Invoice

A purchase receipt in the Chinook dataset.

Target Class: `schema:Invoice`

Table 2: Invoice properties

Description	Path	Type	Cardinality
Customer: An invoice must be linked to exactly one customer.	<code>schema:customer</code>	<code>schema:Person</code>	1..1
Total payment due: An invoice must define a total payment due as a <code>schema:QuantitativeValue</code> .	<code>schema:totalPaymentDue</code>	<code>schema:QuantitativeValue</code>	1..1
Has part: An invoice must have at least one line item (<code>OrderItem</code>).	<code>schema:hasPart</code>	<code>schema:OrderItem</code>	1..*
Date created	<code>schema:dateCreated</code>	<code>xsd:date</code>	0..*

3.3 Music album

A collection of tracks in the Chinook dataset.

Target Class: `schema:MusicAlbum`

Table 3: Music album properties

Description	Path	Type	Cardinality
Name: Every album must have a name.	<code>schema:name</code>	<code>xsd:string</code>	1..1
Artist: Person or music group who created the album.	<code>schema:byArtist</code>	<code>schema:Person</code> or <code>schema:MusicGroup</code>	0..*

3.4 Music Recording

A single music track in the Chinook dataset.

Target Class: `schema:MusicRecording`

Table 4: Music Recording properties

Description	Path	Type	Cardinality
Name: Every track must have a name.	<code>schema:name</code>	<code>xsd:string</code>	1..1
In Album: A track can only belong to a valid <code>schema:MusicAlbum</code> .	<code>schema:inAlbum</code>	<code>schema:MusicAlbum</code>	0..*
Author: The person or group who wrote the track.	<code>schema:author</code>		0..*
Genre	<code>schema:genre</code>	<code>:Genre</code>	1..1
Duration: Track duration must be expressed as a <code>schema:QuantitativeValue</code> .	<code>schema:duration</code>	<code>schema:QuantitativeValue</code>	0..*
Content Size	<code>schema:contentSize</code>	<code>schema:QuantitativeValue</code>	0..*

3.5 Organisation

A company or organization in the Chinook dataset.

Target Class: `schema:Organisation`

Table 5: Organisation properties

Description	Path	Type	Cardinality
Name: Every organisation must have a name.	<code>schema:name</code>	<code>xsd:string</code>	1..1

3.6 Person

Any person (employee, customer, or custom person) present in the dataset.

Target Class: `schema:Person`

Table 6: Person properties

Description	Path	Type	Cardinality
Given Name: Every person must have a given name.	<code>schema:givenName</code>	<code>xsd:string</code>	1..1
Family Name: Every person must have a family name.	<code>schema:familyName</code>	<code>xsd:string</code>	1..1
Email Address: If an email is provided, it must follow a standard email format.	<code>schema:email</code>	<code>xsd:string</code>	0..*
Birth date: A person should have a valid birth date.	<code>schema:birthDate</code>	<code>xsd:date</code>	0..*
Address	<code>schema:address</code>	<code>schema:PostalAddress</code>	0..*
Works for: An employee can report to another person.	<code>schema:worksFor</code>	<code>schema:Person</code> or <code>schema:Organization</code>	0..*
Job title	<code>schema:jobTitle</code>		0..1
knows	<code>schema:knows</code>	<code>schema:Person</code>	0..*

3.7 Quantitative Value

A numerical value with an associated unit.

Target Class: `schema:QuantitativeValue`

Table 7: Quantitative Value properties

Description	Path	Type	Cardinality
Value: A quantitative value must have exactly one numeric value.	<code>schema:value</code>		1..1
Unit Code: A quantitative value must specify its unit via a unitCode URI.	<code>schema:unitCode</code>	<code>sh:IRI</code>	1..1

4 Safety considerations

Information about the explicitly relevant legal bases or a note that during the implementation the relevant legal bases must be observed.

5 Disclaimer

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7 Annex A - References

8 Annex B - Cooperation & Verification

9 Annex C - Abbreviations and Glossary

10 Annex D - Changes in comparison to previous version

11 Annex E - Table of figures

12 Annex F - Table of tables

Bibliography

- [1] R. C. Jackson, J. P. Balhoff, E. Douglass, N. L. Harris, C. J. Mungall, and J. A. Overton, “ROBOT: a tool for automating ontology workflows,” *BMC bioinformatics*, vol. 20, no. 1, p. 407, 2019.
- [2] B. Glimm, I. Horrocks, B. Motik, G. Stoilos, and Z. Wang, “HermiT: an OWL 2 reasoner,” *Journal of automated reasoning*, vol. 53, no. 3, pp. 245–269, 2014.